

1.

→ A solar collector is a device that captures solar radiation & converts it into thermal energy. This heat is transferred to a working fluid like water.

• Working Principle of Solar Collectors:-

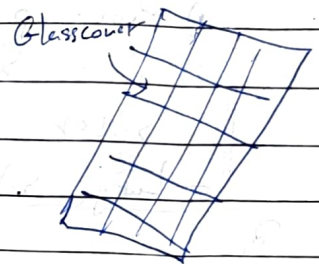
- 1) Sunlight falls on collector surface.
- 2) Absorber plate absorbs radiator.
- 3) Heat is transferred to fluid flowing through tubes.
- 4) Hot fluid is stored or used for heating.

• Types of Solar Collectors:-

A] Non-concentrating collectors

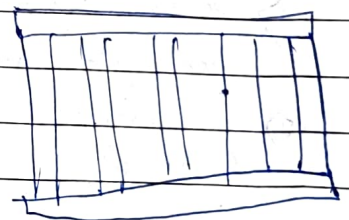
1) Flat Plate collector (FPC)

- Most common type of solar collector.
- Used for domestic hot water.



2) Evacuated tube collector

- Higher efficiency
- Works well in cold climates
- Vacuum tubes reduce heat loss.
- Water gets heated quickly.



B] Concentrating Collectors

1) Parabolic Through Collector

- Cylindrical mirror focuses sunlight onto absorber tube

2) Parabolic Dish Collector.

- Dish mirror focuses sunlight at one point.

3) Central Tower

- Many mirrors reflect sunlight onto receiver.

2.

→ A solar PV system consists of sunlight directly into electricity using PV effect.

Process.

- 1) Sunlight falls on PV cells
- 2) Photons excite electrons in semiconductor
- 3) This creates DC electricity.
- 4) Inverter converts DC → AC power.
- 5) Power is used for loads or fed.
- 6) Batter storage can store extra energy.

Applications

- 1) Rooftop Solar
- 2) Solar Street Light
- Solar Pump for agriculture
- Large Solar

3.

→ Solar thermal tech uses sunlight to generate Heat,

1) Solar Water Heating :-

Used in:- Homes, Motels,

2) Concentrated Solar Power Plants:-

CSP plant generate electricity using steam

3) Industrial Process Heat :-

They require high-temp heat for :-

Food processing, Textile industries, chemical plant.

4) Solar Desalination.

• Converts seawater into fresh water.

5) Solar Cooking

• Cook's food using sunlight converting it into heat.

6) Solar Space Heating :-

• Used in colder countries for room heating.